

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (E&E) (Credit System) Degree Examinations

April 2023

21EE302 - NETWORK ANALYSIS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I

Marks BT* CO* PO*

1. a) Define neutral shift voltage in a 3phase Unbalanced system. Hence derive an expression for Neutral shift voltage in case of a star connected three wire unbalanced load.
- b) Find the voltage across the $5\ \Omega$ resistor in the network shown in Fig Q1B.

5 L*3 2 1,2,5,10

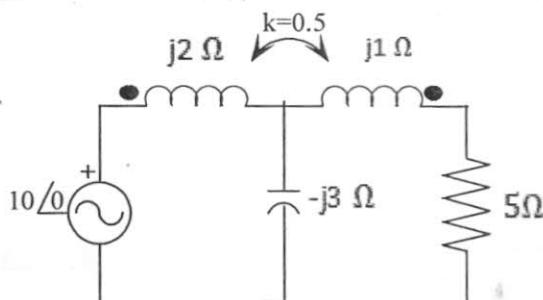


Fig Q1B

7 L3 2 1,2,5,10

- c) A coil of $400\ \Omega$ resistance and $318\ \mu\text{H}$ inductance is connected in parallel with a capacitor 'C' and the circuit resonates at 1MHz . If a second capacitor of $23.5\ \text{pF}$ capacitance is connected in parallel with the first capacitor, find the frequency at which the circuit resonates.

8 L3 21 1,2,5,10

2. a) Define Quality Factor and Resonant frequency of a series RLC circuit. Hence derive an expression for Quality Factor and Resonant frequency.
- b) Determine the power delivered by the voltage source and the current in the $10\ \Omega$ resistor of the network shown in Fig Q2B by loop analysis.

5 L3 2 1,2,5,10

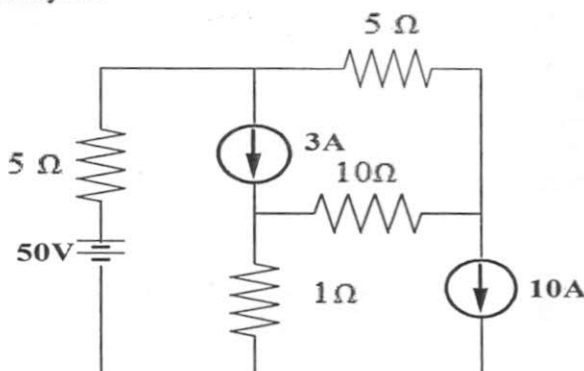


Fig Q2B

7 L3 1 1,2,5,10

- c) A three phase supply of phase sequence ABC, 400V , 4 wire system has a star connected load with $Z_A = 10\ \Omega$, $Z_B = 15 + j10\ \Omega$, $Z_C = j5\ \Omega$. Find the line currents and current through neutral conductor.

8 L3 2 1,2,5,10

3. a) Define coefficient of coupling between two coils. Hence derive an expression for Coefficient of coupling between two coils having Inductances L_1 , L_2 and mutual Inductance M .
- b) An RLC series circuit of 10Ω resistance should be designed to have a bandwidth of 100 Hz. Determine the values of L and C so that circuit resonates at 250 Hz. Also find half power frequencies and Quality factor of the designed circuit.
- c) Find the current through 3Ω resistor in Fig Q3C by node voltage analysis.

5 L2 2 1,2,5,10

7 L3 2 1,2,5,10

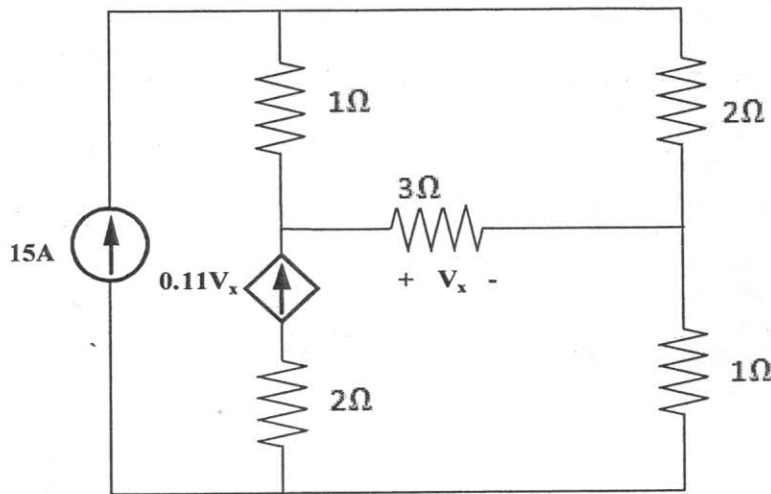


Fig Q3C

8 L3 1 1,2,5,10

Unit – II

4. a) List out the behaviour of circuit elements at (i) $t=0+$ and (ii) $t=\infty$.
- b) Find the voltage v_o across the 2Ω resistor in the Fig Q4B by superposition theorem.

5 L2 4 1,2,4,5,10

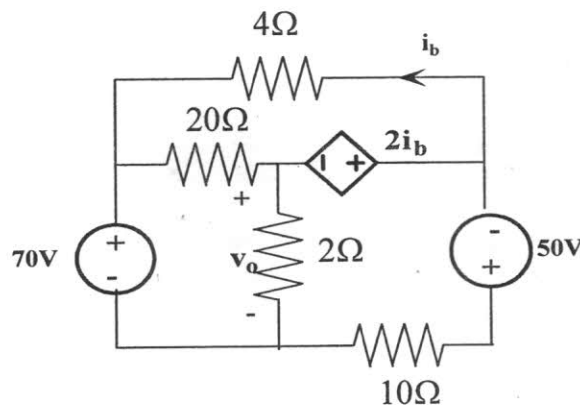


Fig Q4B

7 L3 3 1,2,5,10

- c) At $t=0$, a 5 V pulse of 2sec is applied to a series RL circuit as shown in Fig Q4C. Obtain an expression for $i(t)$.

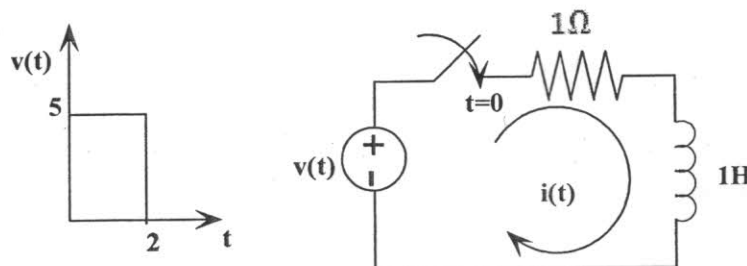


Fig Q4C

8 L4 4 1,2,4,5,10

5. a) Define Millman's theorem. Hence obtain the Millman's network for the Fig Q5A.

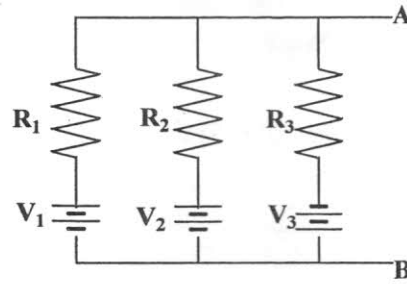


Fig Q5A

5 L3 3 1,2,5,10

- b) In the circuit of Fig Q5B, switch K is moved from position 1 to 2 at $t=0$, Steady condition having reached before switching.

Find i , $\frac{di}{dt}$ and $\frac{d^2i}{dt^2}$ at $t=0+$.

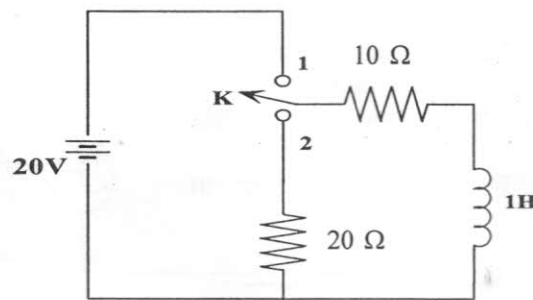


Fig Q5B

7 L3 4 1,2,4,5,10

- c) Determine the current through the 1Ω resistor in the network of Fig Q5C by using Norton's theorem.

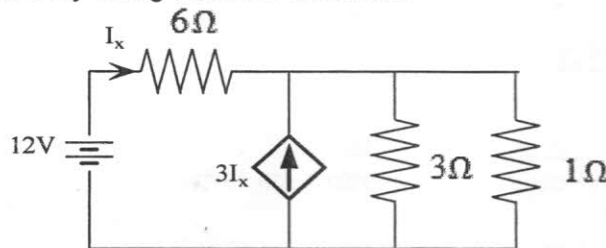


Fig Q5C

8 L4 3 1,2,5,10

6. a) Define time constant of for an RL and RC ac circuit. Hence list out its significance.
b) Determine the current I in the circuit shown in Fig Q6B and then verify the same by Reciprocity theorem.

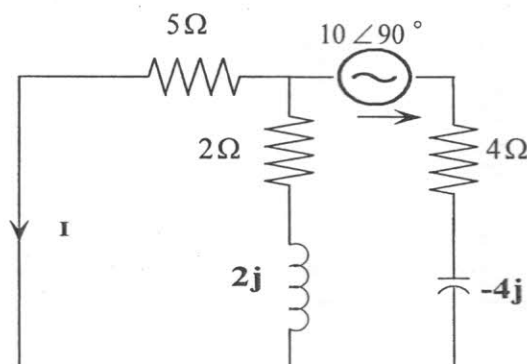


Fig Q6B

5 L2 4 1,2,4,5,10

7 L3 3 1,2,5,10

Find the value of resistance R_L in Fig Q6C for maximum power transfer and calculate the maximum power.

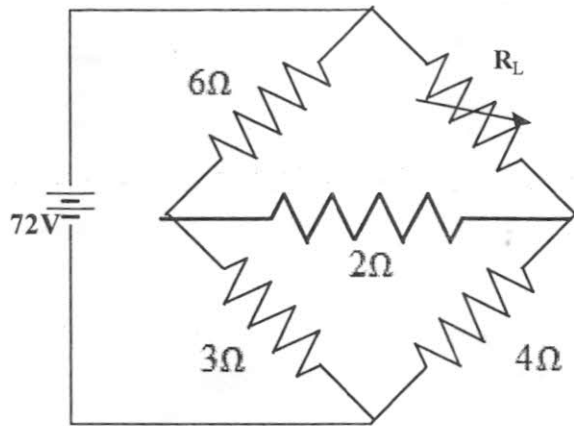


Fig Q6C

8 L4 3 1,2,5,10

Unit – III

7. a) Obtain an expression for the current flowing in RC network supplied by a voltage source of $V_u(t)$ using Laplace transform method. Also find the expression for voltage across the capacitance.
b) Determine the y parameters for the network shown in Fig Q7B.

10 L3 5 1,2,5,10

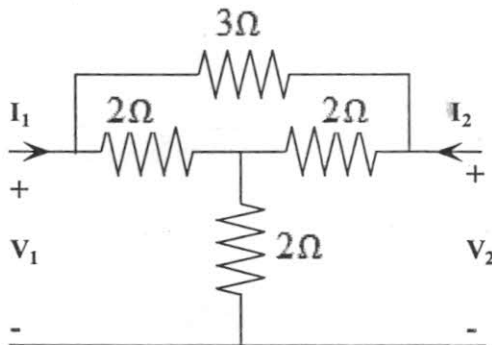


Fig Q7B

10 L3 5 1,2,5,10

8. a) Define hybrid parameters and z parameters. Hence Obtain hybrid parameters in terms of z parameters.
b) In the network of Fig Q8B, the switch is closed at $t=0$ with the network being previously unenergized. Determine the expression for current $i_1(t)$ using Laplace Transform method.

10 L3 5 1,2,5,10

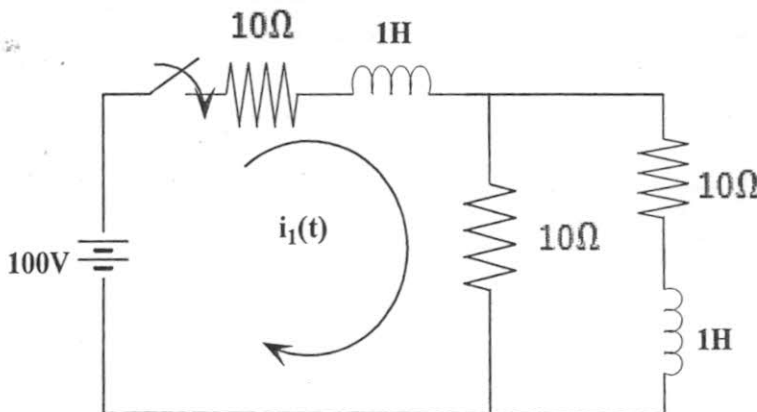


Fig Q8B

10 L3 5 1,2,5,10

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
